

General Directions

1. Use legible/clear handwriting
2. For your answers use the separate answer sheet provided.
3. Use uppercase letters for your answers (for Parts I through V).
4. Never use pencils to write your answers. Corrections are forbidden for any work done with a pencil.
5. Make sure that this exam consists of EIGHT printed pages.

Part I. Write 'TRUE' if the statement is correct or 'FALSE' if it is incorrect and write down your answer on the separate answer sheet provided.

1. Eukaryotic cells have DNA that is in a continuous loop.
2. Chemiosmosis is the diffusion of proton across ATP synthase.
3. Robert Hooke proposed that cells form by free-cell formation.
4. The archaebacteria are thought to be the oldest organisms on Earth.
5. The nature of the fatty acids influences the fluidity of the membrane.
6. Pure, liquid water has a higher water potential than any other system.
7. Plants are green because they absorb green and reflect other wavelength.
8. Photolysis is the splitting of water molecules with the help of chemical energy.
9. The process of adding phosphate group to a molecule is known as phosphorylation.
10. The molecules of the photosystems and the electron transport chains are fixed in thylakoid membranes.

Part II. Match the premises under the column "A" with the responses given under the column "B" by writing the letter of your choice on the separate answer sheet provided.

Column 'A'

11. They have their own DNA
12. They digest worn out organelles
13. The subunits that are manufactured in the nucleolus.
14. They are the site of the light-dependent reactions of photosynthesis.
15. They provide temporary storage of water, food, and waste products.

Column 'B'

- A. Vacuoles
- B. Thylakoids
- C. Ribosomes
- D. Lysosomes
- E. Mitochondria
- F. Golgi apparatus
- G. Endoplasmic reticulum

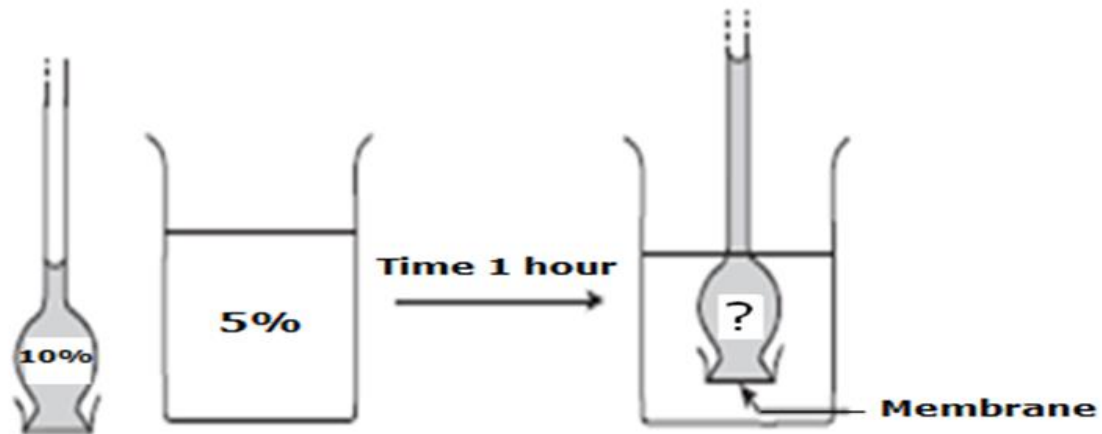
Part III. Fill in the blank spaces with the most appropriate words on the separate answer sheet provided.

16. The cluster of pigment molecules is called _____.
17. Oxygen-dependent production of ATP is known as _____.
18. An instrument for measuring respiration is known as _____.
19. The theory of the origin of eukaryotic cells is called the _____.
20. The electrons lost by chlorophyll in photosystem II are replaced by _____.
21. The individual structures in a cell with a specific function is said to be _____.
22. The integral proteins that move medium sized particles across the membrane are called _____.
23. The proteins used to actively transport substances across plasma membranes are known as _____.
24. The processes which rely on the kinetic energy of the particles of substances and on concentration gradients are known as _____.
25. The first molecule to accept the excited electron displaced from a chlorophyll molecule is called _____.

Part IV. Choose the correct answer from the alternatives given and write the letter of your choice on the separate answer sheet provided.

26. What is the sequence of organelles that a secreted protein would have passed through on its journey out of a cell?
- A. Mitochondria, Golgi apparatus, cell membrane, cytoplasm
 - B. Cell membrane, mitochondria, Golgi apparatus, cytoplasm
 - C. Golgi apparatus, rough endoplasmic reticulum, cell membrane
 - D. Rough endoplasmic reticulum, Golgi apparatus, cell membrane
27. What did Anton van Leeuwenhoek call when he first observed moving unicellular organisms (protocists) under his simple microscope?
- A. Cellula
 - B. Bacteria
 - C. Animalcules
 - D. Schwann cells
28. Which model of membrane structure is the oldest?
- A. Fluid-mosaic model
 - B. Unit membrane model
 - C. Davson-Danielli model
 - D. Singer-Nicholson model
29. Which statement would be included in a summary of the cell theory? All cells
- A. come from other, preexisting cells.
 - B. are made up of at most a million cells.
 - C. contain a nuclear envelope to protect the nucleus.
 - D. are the basic unit of structure except for bacteria.
30. Which process do some animal cells use to engulf, digest, and destroy invading bacteria?
- A. Exocytosis
 - B. Pinocytosis
 - C. Endosmosis
 - D. Phagocytosis
31. In receptor-mediated endocytosis, receptor molecules initially project to the outside of the cell. Where do they end up after endocytosis? On the
- A. outside surface of vesicles.
 - B. inside surface of the vesicle.
 - C. outer surface of the nucleus.
 - D. inside surface of the nucleus.
32. Which of the following organelle and its cell type matched correctly?
- A. Prokaryotes – have no distinct nucleus
 - B. Prokaryotes – have no genetic material
 - C. Eukaryotes – have small number of ribosomes
 - D. Eukaryotes – have membrane bonded ribosomes

33. In an experiment, the bottom of a thistle tube is covered with a membrane permeable only to water. The thistle tube is filled with a 10% sucrose solution and placed in a beaker containing a 5% sucrose solution.



- What will happen to the solution in the tube after one hour?
- A. The concentration of sucrose will increase and become isotonic to the solution in the beaker.
 - B. The concentration of sucrose will decrease and become isotonic to the solution in the beaker.
 - C. The concentration of sucrose will decrease and become hypotonic to the solution in the beaker.
 - D. The concentration of sucrose will increase and become hypotonic to the solution in the beaker.
34. The theory of spontaneous generation states that all life on earth originates from
- A. prokaryotes.
 - B. living things.
 - C. celestial body.
 - D. inanimate matter.
35. Which one of the following events occurred last in the cell biology?
- A. Describing the Golgi apparatus in cells
 - B. Building the first compound microscope
 - C. Drafting the human genome DNA sequence
 - D. Proposing the double-helix structure of DNA

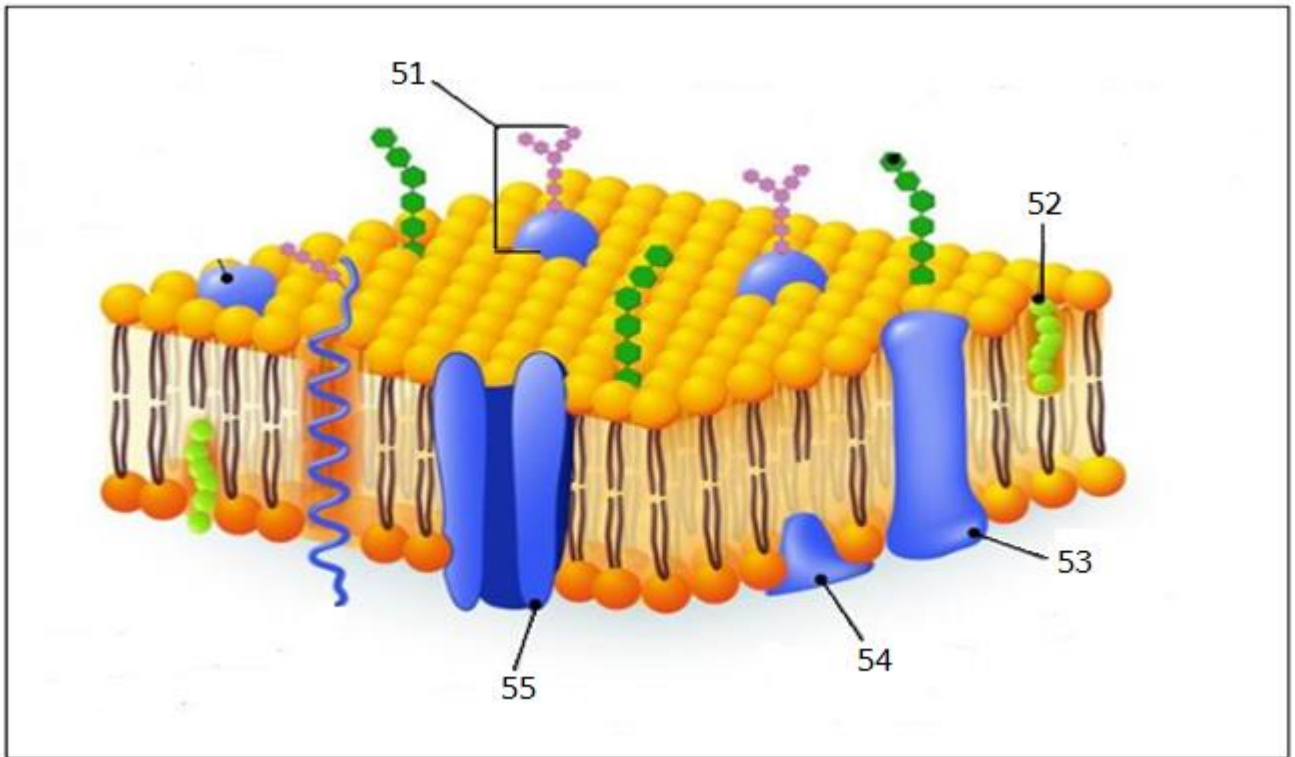
-
36. Cube A has a side measuring 4 mm. Cube B has a side measuring 12 mm. What is the surface-area-to-volume ratio of cube B when compared to cube A? It is
- A. four times bigger. C. three times bigger.
B. four times smaller. D. three times smaller.
37. Why is the surface-area-to-volume ratio of a cell so important? Because it is a measure of how the cell
- A. releases efficiently energy in cellular respiration.
B. uses efficiently the energy it releases in respiration.
C. is efficient enough in conserving and storing energy.
D. is efficient in gaining the oxygen it needs for respiration.
38. What is the size of the smallest bacterial cell in length?
- A. 0.1 μm C. 10 μm
B. 1.0 μm D. 100 μm
39. Which biologist first stated the cell theory?
- A. Rudolf Virchow C. Theodor Schwann
B. Rene Dutrochet D. Matthias Schleiden
40. Robert Hooke, English scientist, first coined the word 'cell' when he examined
- A. dead cells in cork tissue.
B. dead cells in skin tissue.
C. living cells in skin tissue.
D. living cells in cork tissue.
41. The number of ATP molecules produced from the complete respiration of one molecule of glucose to carbon dioxide and water is
- A. 2 C. 24
B. 12 D. 36
42. In yeast cells,
- A. alcohol is produced after glycolysis.
B. lactic acid is produced during glycolysis.
C. alcohol is produced during the citric acid cycle.
D. lactic acid is produced during anaerobic respiration.
43. Where do the H^+ ions go once the H_2O molecule is separated?
- A. Stoma C. Cytoplasm
B. Stroma D. Thylakoid space
44. Which of the following are products of the light reactions of photosynthesis that are utilized in the Calvin cycle?
- A. H_2O and O_2 C. CO_2 and glucose
B. ATP and NADPH D. ADP, Pi , and NADP

45. What part of the chloroplast does the Calvin cycle take place?
A. Stroma
B. Thylakoid
C. Inner membrane
D. Outer membrane
46. Which enzyme is responsible for carbon fixation in C₃ photosynthesis?
A. Rubisco
B. Hexokinase
C. Dehydrogenase
D. PEP carboxylase
47. Which of these would be considered a C₄ pathway plant?
A. Algae
B. Cactus
C. Oak tree
D. Sugar cane
48. The first stable product of dark reaction in C₄ plants
A. glycolic acid.
B. oxaloacetic acid.
C. phosphoglycolic acid.
D. phosphoglyceric acid.
49. The three circles represent three major processes in aerobic respiration.



- Process 'C' represents
- A. glycolysis.
B. krebs cycle.
C. electron transport chain.
D. substrate level phosphorylation.
50. The ATP produced during glycolysis is generated by which of the following?
A. Chemiosmosis
B. Electron transport chain
C. Oxidative phosphorylation
D. Substrate level phosphorylation

Part V. The diagram below illustrates the cross section of a leaf. Identify the parts designated by the numbers 51-55 and write down your answer on the separate answer sheet provided.



Part VI. Read the following questions carefully and write down brief answers on the separate answer sheet provided.

56. Distinguish between prokaryotic and eukaryotic cells.
57. What accounts for the fluidity of the fluid-mosaic model of membrane structure?
58. What is the significance of inward convolutions or folds of the inner membranes of mitochondria?
59. What requirements should the particles meet in order to pass through the plasma membrane by simple diffusion?
60. Suppose 80 divisions on the scale of an eyepiece graticule correspond to 32 small divisions on the stage micrometer. Each small division on the stage micrometer = 20 μm . 8 cells fit across 80 divisions of the eyepiece graticule. Calculate the length of each cell.